Point-by-Point Response

Comment 1 — Clarification of the methodological contribution and novelty

Comment:
The manuscript should clarify what CTD contributes beyond conventional ablation studies, sensitivity analysis, factorial/DOE methods, and ranking-stability analysis. The current presentation does not sufficiently establish the methodological novelty.

Response

I thank the Editor for identifying this important issue. I agree that the original manuscript did not sufficiently distinguish CTD from established experimental-analysis approaches.

I have therefore substantially revised the methodological positioning of CTD.

In the revised manuscript, CTD is explicitly presented as a **complementary counterfactual attribution framework**, rather than as a replacement for conventional ablation, sensitivity analysis, factorial/DOE methods, or ranking-stability analysis.

The revised manuscript now distinguishes the approaches as follows:

- conventional ablation evaluates the effect of removing or changing components, generally through one-factor-at-a-time comparisons;
- sensitivity analysis characterizes response variation across selected parameter or decision changes;
- factorial/DOE approaches provide systematic estimation of main and interaction effects when a factorial design is explicitly specified;
- ranking-stability analysis evaluates whether model ordering changes under perturbations;
- CTD combines controlled counterfactual intervention, effect attribution, conclusion-fragility analysis, and selected interaction analysis within a common experimental tuple.

I have therefore revised the claims of novelty so that CTD is not characterized as replacing these established methodologies. Instead, its contribution is framed as an **integrated experimental attribution perspective for studying how non-algorithmic experimental decisions affect ML conclusions**.

The revised manuscript also includes explicit methodological comparisons with ablation, sensitivity analysis, and factorial/DOE approaches.

Location in revised manuscript: Methodology and methodological-positioning discussion, including Section 5.7.

Comment 2 — Limited empirical generalization

Comment:
The original empirical validation used only three datasets, four classical models, three seeds, and one interaction pair. The revised study should provide broader empirical validation.

Response

I agree.

The original evaluation was expanded substantially. The revised production analysis uses:

- **five datasets:** Breast Cancer Wisconsin, Bank Marketing, Pima Indians Diabetes, Spambase, and Phoneme;
- **four classical ML models:** Logistic Regression, Decision Tree, Random Forest, and Linear SVM;
- **fifteen prespecified random seeds**;
- the same controlled counterfactual framework across the production datasets.

This provides a substantially broader evidence base than the original three-dataset, three-seed analysis.

The resulting main evidence base contains **2,100 audited rows**, comprising the baseline, single-decision counterfactuals, and completed joint-intervention conditions.

I also explicitly acknowledge that the revised study remains a benchmark-based empirical evaluation rather than an exhaustive evaluation across all possible datasets, algorithms, and application domains.

Location: Experimental design, Results, and Limitations sections.

Comment 3 — Increase the number of repeated runs/seeds

Comment:
Only three random seeds were used originally. More repeated runs are required to establish the stability of the observed effects.

Response

I agree.

The revised study uses **15 prespecified random seeds**:

42, 123, 2026, 7, 19, 31, 53, 71, 89, 107, 149, 173, 211, 257, and 311.

The same seed set is used consistently within the production analysis. The counterfactual configurations reuse the corresponding dataset-seed split so that comparisons are not confounded by independently changing the evaluation partition.

This substantially increases the repeated-run evidence relative to the original three-seed design.

I also report distributional statistics and bootstrap uncertainty intervals for the completed interaction effects rather than relying solely on individual runs.

Location: Experimental Protocol and Results Sections.

Comment 4 — The original No Scaling × PCA95 interaction is intrinsically dependent

Comment:
The original interaction between No Scaling and PCA95 is intrinsically dependent because PCA is sensitive to scaling. This interaction therefore provides a limited basis for demonstrating general interaction effects.

Response

I agree with this concern.

Rather than treating the original **No Scaling × PCA95** interaction as the principal demonstration of CTD, I redesigned the interaction analysis around less intrinsically dependent experimental decisions.

The revised experimental protocol considered the following candidate interaction families:

1. **Scaling × Training Fraction**
2. **Sampling × Training Fraction**
3. **Sampling × Representation**
4. **Representation × Training Fraction**

The production interaction analysis completed in the current revision focuses on the two interaction families for which the revised production evidence was successfully completed:

- Scaling × Training Fraction
- Sampling × Training Fraction

The original No Scaling × PCA95 interaction is therefore not used as the flagship empirical interaction claim.

This change directly addresses the Editor's concern regarding the dependence between scaling and PCA.

I have also explicitly discussed this limitation and the scope of the completed interaction analysis rather than presenting the revised interaction evidence as exhaustive.

Location: Interaction methodology, Results Sections 5.2–5.6, and Limitations.

Comment 5 — Multiple interaction pairs should be evaluated

Comment:
The manuscript should evaluate several interaction pairs rather than relying on one interaction.

Response

I agree.

The revised experimental protocol was designed to evaluate multiple theoretically meaningful interaction families. The completed production evidence contains two joint-intervention families:

- Scaling × Training Fraction
- Sampling × Training Fraction

For each family, **300 interaction estimates** were obtained across the five production datasets, four models, and fifteen seeds, resulting in **600 interaction estimates** in total.

The revised manuscript reports their distributions, uncertainty, threshold-based magnitudes, dataset-level heterogeneity, and effects on model ranking.

I have deliberately avoided presenting the two completed interaction families as an exhaustive enumeration of all possible interactions. Additional interaction families remain an explicit direction for future work.

Location: Interaction Analysis and Results Sections 5.2–5.7.

Comment 6 — Ranking reversals need uncertainty and practical-effect thresholds

Comment:
Ranking reversals should not be treated simply as binary events. The manuscript should consider uncertainty and practical-effect thresholds.

Response

I thank the Editor for this important suggestion and have incorporated it directly.

The revised analysis distinguishes between:

1. raw ranking changes;
2. pairwise ranking reversals; and
3. ranking reversals that exceed prespecified practical-effect thresholds.

Three practical-effect thresholds were specified:

- $|I| \geq 0.005$
- $|I| \geq 0.010$
- $|I| \geq 0.020$

For the Sampling × Training interaction, the proportions exceeding these thresholds are:

- 66.7% at 0.005;
- 36.3% at 0.010;

- 22.0% at 0.020.

For Scaling × Training:

- 30.0% at 0.005;
- 18.0% at 0.010;
- 9.3% at 0.020.

I also report bootstrap 95% confidence intervals for the aggregate interaction effects.

For ranking analysis, Sampling × Training produced **134/300 (44.67%)** model-ranking changes and **110/450 (24.44%)** pairwise reversals. Scaling × Training produced **105/300 (35.00%)** ranking changes and **73/450 (16.22%)** pairwise reversals.

The revised manuscript explicitly states that the thresholds are **practical-effect criteria rather than statistical significance thresholds**.

Location: Sections 5.2, 5.4, and 5.5.

Comment 7 — Need comparison with standard sensitivity/factorial methods

Comment:
The manuscript should explicitly compare CTD with conventional sensitivity and factorial/DOE approaches.

Response

I agree.

A new methodological-positioning discussion has been added to the revised manuscript.

The discussion explains that CTD shares important principles with factorial experimentation and sensitivity analysis, particularly controlled variation of experimental factors. However, CTD emphasizes the attribution of observed ML performance differences to explicit counterfactual changes in experimental decisions and connects those effects to downstream conclusions such as model ranking.

I therefore present CTD as **complementary to established experimental-design methods**, rather than claiming methodological replacement.

This revised framing substantially narrows the original novelty claim and clarifies the intended contribution.

Location: Section 5.7 and revised methodological discussion.

Comment 8 — Reproducibility and public availability

Comment:
The revised study should provide public code/configurations and sufficient information for reproduction.

Response

I agree.

The revised experimental protocol has been substantially strengthened to document:

- dataset identities;
- dataset characteristics;
- preprocessing decisions;
- model configurations;
- random seeds;
- train/test split construction;
- counterfactual configurations;
- interaction definitions;
- evaluation metrics;
- practical-effect thresholds;
- run accounting;
- validation procedures; and
- evidence-ledger structure.

The production analyses were executed with explicit manifests, run ledgers, checkpointing, and validation gates.

The revised manuscript also strengthens the reproducibility statement and identifies the code/configuration materials associated with the experimental protocol.

Location: Experimental Protocol, Reproducibility Statement, and Data/Code Availability sections.

Comment 9 — Accuracy of reported results

Comment:
All revised numerical results should be accurate and internally consistent.

Response

I agree completely.

The revised results were rebuilt from the audited production evidence rather than manually modifying the original results.

For the completed joint-intervention analysis:

- 600 new joint-condition rows were produced;
- 1,500 base/single-condition rows were reused;
- the combined evidence base contains 2,100 rows;
- 600 interaction values were independently recomputed;
- the maximum interaction-arithmetic discrepancy was effectively zero.

The figure-generation pipeline independently reconstructed the interaction values and pairwise comparisons before producing the revised empirical figures.

This procedure was used to reduce the possibility of transcription or calculation errors between the production ledgers and the manuscript.

Comment 10 — Revised figures

Comment:
The revised manuscript should provide clear empirical visualization consistent with the expanded analysis.

Response

I have regenerated the empirical figures from the revised audited evidence.

Figure 1 retains the CTD framework.

The superseded empirical figures from the original three-dataset, three-seed analysis were not carried forward because they represented the earlier experimental design.

The revised manuscript now contains:

- **Figure 2:** Interaction distributions;
- **Figure 3:** Dataset-level heterogeneity;
- **Figure 4:** Model-ranking reversals;
- **Figure 5:** Practical-effect reversals.

The corresponding captions and in-text references have also been revised.

The four empirical figures were generated at publication-oriented high resolution and are supplied separately as PNG files in addition to being embedded in the revised manuscript.

Comment 11 — Tone down conclusions and claims of generality

Comment:
The conclusions should be less overstated and should distinguish empirical findings from broader claims.

Response

I agree.

The conclusion has been rewritten to avoid implying that the observed effects establish universal behavior across machine-learning systems.

The revised manuscript now states the findings in terms of the evaluated benchmark, models, experimental decisions, and prespecified protocol.

I explicitly distinguish:

- empirical findings supported by the five-dataset study;
- methodological interpretation;
- practical implications; and
- unresolved generalization questions.

The revised manuscript therefore does not claim that every experimental decision necessarily produces ranking instability or that CTD universally dominates existing experimental-analysis methods.

Comment 12 — Explicit limitations

Comment:
The manuscript should provide a clearer discussion of limitations.

Response

A substantially expanded limitations discussion has been added.

The revised manuscript explicitly acknowledges:

1. the benchmark scope of five production datasets;
2. the use of four classical ML models;
3. the fact that Covertype was retained as a resource-validation case rather than included in the final production analysis;
4. the fixed stratified evaluation protocol;
5. the two completed production interaction families;
6. the fact that practical-effect thresholds are interpretive criteria rather than significance tests;
7. the benchmark-specific nature of ranking-reversal frequencies; and

8. the need for additional datasets, algorithms, domains, and interaction families in future research.

These limitations are now incorporated directly into the interpretation of the findings rather than being presented only as a final caveat.

Response concerning the original empirical figures

Because the revised experimental design supersedes the original three-dataset/three-seed empirical analysis, I did **not** carry the old empirical Figures 2–5 into the revised manuscript.

The original CTD framework figure was retained.

The empirical figures were regenerated from the expanded evidence base so that the figures, tables, numerical results, and conclusions all correspond to the revised experimental protocol.

This also avoids presenting results from the superseded No Scaling × PCA95 interaction as though they were part of the final expanded study.

Closing Response

I sincerely appreciate the Editor and Reviewers for the constructive comments. The revision has resulted in substantial changes to the experimental design, methodological positioning, empirical validation, interaction analysis, uncertainty reporting, ranking analysis, figures, reproducibility documentation, and limitations.

I believe the revised manuscript provides a more carefully scoped and empirically supported evaluation of CTD while avoiding claims that extend beyond the evidence.

I thank the Editor and Reviewers for the opportunity to substantially improve the manuscript.